

'CONTROL+ALT+DELETE': E-WASTE MANAGEMENT IN CHENNAI

Approaches, Challenges, and Consumer Behaviour

Authors: 1. SHRIRAM. N (II MA ECONOMICS), DG Vaishnav College, Chennai.

2. ARAVIND RAJ. G (II MA ECONOMICS), DG Vaishnav College, Chennai.

Contact details: Email ID: sriram.narasiman58@gmail.com | aravindgrj@gmail.com

Mobile number: +91 9566242049 | +91 9884463204

ABSTRACT:

'Electronic waste', commonly referred to as **'e-waste'**, has become a critical environmental issue due to the rapid growth of technology and consumer electronics. Cities like Chennai, with a high rate of technological adoption, generate substantial amounts of e-waste, leading to serious concerns regarding environmental sustainability and resource depletion. **Sustainable development** emphasizes the responsible management of e-waste to balance economic growth with environmental protection, ensuring that future generations are not burdened by the consequences of improper disposal.

This study explores e-waste disposal behavior and awareness in Chennai, analyzing how individuals and households manage their electronic waste. Findings indicate that while a significant portion of the population is aware of e-waste, many lack sufficient knowledge about its environmental hazards and proper disposal methods. Common disposal practices include storing old electronics at home, selling them to informal scrap dealers, or discarding them with regular waste, contributing to environmental pollution and inefficient resource recovery.

E-waste management in Chennai faces multiple challenges, including inadequate collection infrastructure, limited formal recycling facilities, and the prevalence of informal e-waste handlers who do not follow environmentally safe disposal methods. The presence of hazardous substances such as lead, mercury, and cadmium in electronic waste poses severe risks to human health and the ecosystem. Additionally, valuable materials such as gold, silver, and rare earth metals often go unrecovered due to ineffective recycling practices.

Understanding public behavior and awareness regarding e-waste disposal is crucial in addressing the broader implications of e-waste management on sustainable development. With the increasing generation of e-waste in Chennai, analyzing current disposal patterns provides valuable insights into the environmental and economic impact of electronic waste, highlighting the urgent need for improved management strategies.

Keywords: Electronic waste, Sustainable Development, Recycle, E-waste Management, Pollution, Consumer Awareness, Refurbished electronics, Extended Producer Responsibility.

INTRODUCTION:

Sustainable development is a widely acknowledged concept that seeks to harmonize economic growth, environmental preservation, and social well-being, ensuring that present needs are met without jeopardizing the ability of future generations to meet theirs. This comprehensive approach integrates environmental sustainability, economic advancement, and social equity, making it a vital framework for long-term global stability.

The significance of sustainable development gained momentum with the release of the Brundtland Report (1987), which defined it as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs." This principle underscores the responsible utilization of resources, pollution control, and environmental conservation while simultaneously fostering economic progress and enhancing quality of life.

Sustainable development is built upon three fundamental pillars:

- **Environmental Sustainability** – Advocating for the responsible consumption of natural resources, minimizing pollution, and combating climate change to safeguard ecosystems.
- **Economic Sustainability** – Encouraging economic expansion and innovation while preserving ecological balance and preventing resource depletion.
- **Social Sustainability** – Promoting inclusivity, equity, and access to essential services such as education, healthcare, and clean water for all individuals.

The urgency for sustainable development has never been greater, as challenges such as rapid industrialization, population growth, and ecological degradation threaten the planet's stability. Achieving sustainability necessitates collaborative efforts from governments, businesses, and individuals to adopt policies and practices that support environmental conservation, economic resilience, and social justice.

E waste management comes under the first pillar, that is Environmental Sustainability. Electronic waste has become a significant public health and environmental concern in India. The country ranks as the third-largest producer of e-waste globally, generating approximately 2 million tons of electronic waste annually, with an additional undisclosed volume being imported from other nations.

According to a report by the United Nations Trade and Development (UNCTAD), India experienced the highest global growth rate of 163% in e-waste generation from screens, computers, and small IT and telecommunication

equipment (SCSIT) between 2010 and 2022. This rapid increase highlights the urgent need for efficient e-waste management and sustainable disposal practices.

Each year, computer devices contribute to nearly 70% of total e-waste, while the telecom sector accounts for 12%, medical equipment for 8%, and electrical equipment for 7%. The majority of e-waste, approximately 75%, is generated by the government, public sector, and private sector companies, whereas individual households contribute only 16%.

OBJECTIVES:

The rapid expansion of technology and increased reliance on electronic devices have led to a significant rise in electronic waste (e-waste). Managing this waste effectively is crucial to prevent environmental pollution and health hazards. Chennai, as a major urban center, faces challenges in handling e-waste due to inadequate disposal practices, informal recycling, and limited public awareness.

This research aims to examine the current e-waste management landscape in Chennai, focusing on existing policies, infrastructure, and community participation. By identifying challenges and evaluating potential improvements, the study will contribute to developing more effective and sustainable e-waste management strategies. The following objectives outline the key areas of focus for this study:

- To analyse the knowledge, attitude and practice of e-waste management among the stakeholders in Chennai.
- To determine which type of gadgets get disposed of frequently.
- To identify Environmental and Health Impacts.
- To suggest Sustainable Solutions and Best Practices.
- To assess the Current E-Waste Generation and Trends.

HYPOTHESIS:

H₀: There is no significant relationship between income level and the frequency of purchasing new electronic products.

H₀: The method of e-waste disposal (trash, scrap, or recycling) is independent of the level of education of the consumer.

RESEARCH METHODOLOGY:

The study adopts a quantitative research approach to analyze e-waste management practices in Chennai. A structured questionnaire was used to collect data from respondents, focusing on their awareness, disposal methods, and opinions on e-waste management policies.

A **Convenience Random Sampling Method** was employed to select respondents for the study. This method was chosen to ensure accessibility and ease of data collection while capturing diverse perspectives from households, businesses, and other stakeholders involved in e-waste management. In total, over **100 responses** were collected from participants residing or operating within Chennai district.

The research focuses on **Chennai district**, a major metropolitan area with high electronic consumption and waste generation. The study covers various regions within the district to ensure a comprehensive analysis of e-waste management practices.

For analysis, some of the popular statistical tools like **percentage analysis** and **correlation** were incorporated into the study.

These analytical methods help in interpreting the collected data effectively and drawing conclusions that contribute to improved e-waste management strategies in Chennai.

LITERATURE REVIEW:

RAJIV GANGULY (2016) in the article named “**E-WASTE MANAGEMENT IN INDIA – AN OVERVIEW**” highlights the growing complexity of solid waste management in India due to e-waste. It examines e-waste generation, disposal practices, and recycling sectors. Sustainable strategies, stakeholder roles, policy frameworks, and producer responsibility are discussed. effective e-waste management requires integrated policies, technological solutions, and public awareness to minimize environmental hazards and promote sustainable waste practices.

DR. NEHA GARG, DEEPAK KUMAR ADHANA (2019) in the article named” **E-WASTE MANAGEMENT IN INDIA: A STUDY OF CURRENT SCENARIO**” emphasizes responsible e-waste disposal by selecting certified recyclers and utilizing retailer take-back programs. The CPCB provides a list of authorized recyclers, and companies like Samsung promote recycling initiatives. Conclusively, consumer awareness, responsible purchasing, and community engagement are essential for sustainable e-waste management, ensuring environmental protection for future generations.

Mahesh C. Vats , Santosh K. Singh (2014) in the journal named “**STATUS OF E-WASTE IN INDIA - A REVIEW**” suggest integrating user awareness, efficient collection, and controlled treatment, alongside PPP and FDI-driven infrastructure, can enhance India’s e-waste management, ensuring sustainability and public health.

Muskan Jain , Depak Kumar , Jyoti Chaudhary , Sudesh Kumar , Sheetal Sharma , Ajay Singh Verma (2023) in a journal named “**REVIEW ON E-WASTE MANAGEMENT AND ITS IMPACT ON THE ENVIRONMENT AND SOCIETY**” highlights the dominance of the informal sector in e-waste management, posing significant environmental and health risks, especially in developing nations. Despite regulations, formal recycling systems remain ineffective. However, adopting advanced life cycle assessment models, prioritizing vulnerable groups, and fostering global collaboration are essential for mitigating e-waste exposure and health hazards.

Vijay Kumar Garlapati (2016) in the article named “**E-WASTE IN INDIA AND DEVELOPED COUNTRIES: MANAGEMENT, RECYCLING, BUSINESS AND BIOTECHNOLOGICAL INITIATIVES**” pointed out that a rapid growth of e-waste, initially emerging in developed nations and now affecting developing countries like India. Rapid technological advancements accelerate obsolescence, increasing waste generation. Establishing an efficient recycling system requires precise classification and quantification of valuable and hazardous components, ensuring economic feasibility and environmental sustainability.

R Ganesan, B Ramesh, and Charan Teja (2020) in the Journal named “**AWARENESS OF REUSE REDUCE RECYCLE & DISPOSE OF E-WASTE IN CHENNAI**” environmental and health hazards of e-waste disposal, particularly incineration and landfilling, which contribute to air pollution, soil degradation, and acidification. Raising awareness, implementing safe recycling practices, and promoting eco-friendly materials are essential for achieving sustainable waste management and a zero e-waste.

Dr.A.Sivagami (2020), “**AWARENESS OF INDIVIDUAL CONSUMERS TOWARDS E-WASTE MANAGEMENT IN CHENNAI CITY, TAMIL NADU**”. The study highlights the importance of consumer awareness in efficient e-waste management. Research conducted in Chennai revealed that only 32% of consumers were aware of proper disposal methods. Thus, strengthening consumer education and engaging various stakeholders, including formal and informal sectors, is crucial for enhancing e-waste management strategies and sustainability.

E-WASTE MANAGEMENT IN INDIA:

India recycled only **32.9% of the e-waste** generated in **2021-22**, leaving a **staggering 67% (10,74,024 tonnes) unprocessed**. While the recycling rate has improved over the years, a significant amount of e-waste remains unmanaged, posing serious environmental and health risks. Toxic substances such as **lead, cadmium, mercury, arsenic, and asbestos** present in e-waste can leach into the soil and water if not disposed of scientifically. Additionally, much of this untreated waste is handled by **informal industries**, where children are often exposed to hazardous chemicals, affecting their growth and development.

India has witnessed a **steady rise in e-waste recycling**, with rates increasing from **9.79% in 2017-18 to 32.9% in 2021-22**. Despite this progress, a large portion of e-waste is still **processed informally**, as formal recycling facilities remain **underutilized**. In 2021-22, **Haryana** led in e-waste collection and processing, handling **2,45,015.82 tonnes**. However, the government's official data did not account for all states and Union Territories, indicating possible gaps in reporting.

India is the **largest e-waste generator in South Asia** and the only country in the region with **dedicated e-waste management laws**. However, **90% of the country's e-waste is managed by the informal sector**, which lacks proper safety measures and regulations. Various agencies, including the **United Nations and the Centre for Science and Environment**, have stressed the need to regulate this sector to ensure responsible recycling. The government's

E-Waste (Management) Rules 2022, replaced the 2016 regulations, aim to **address these challenges** by mandating **registration of manufacturers, producers, refurbishers, and recyclers** on an online portal developed by the **Central Pollution Control Board**.

The new rules, effective **April 1, 2023**, also expand the scope of regulation by **increasing the number of covered electrical and electronic items from 21 to 106**. They introduce **restrictions on hazardous substances** in electronic products and emphasize **worker safety, skill development, and monitoring** in the dismantling and recycling sectors. While these regulations mark a significant step forward, their success will depend on **strict enforcement, better public awareness, and the integration of informal recyclers into the formal system**.

- Harmful effects of unsafe disposal of e-wastes include:
 - **Soil Degradation**
Toxic components in e-waste, such as **lead, cadmium, and arsenic**, can seep into the soil, reducing its fertility and making it unfit for agriculture.
 - **Water Contamination**
Hazardous chemicals from discarded electronics **leach into groundwater and nearby water bodies**, polluting drinking water and endangering aquatic ecosystems.
 - **Air Pollution**
Burning e-waste releases harmful gases like **dioxins and furans**, which contribute to air pollution and pose serious respiratory health risks.
 - **Human Health Risks**
Exposure to e-waste toxins can lead to **neurological disorders, kidney damage, respiratory problems, and even cancer**.
 - Children and pregnant women face a higher risk of **developmental complications** due to prolonged exposure.
 - **Harm to Wildlife**
When toxic substances enter food chains, they **adversely affect animals, birds, and marine life**, leading to biodiversity loss.
 - **Hazards in Informal Recycling**
Many workers in **unregulated e-waste processing** handle hazardous materials **without protective gear**, leading to long-term health problems.
 - **Contribution to Climate Change**
Improper disposal and incineration of electronic waste release greenhouse gases, which contribute to **global warming**.
 - **Wastage of Valuable Resources**

E-waste contains reusable materials like **gold, silver, and copper**, which are lost when not properly recycled.

- **Risk of Fire and Explosions**

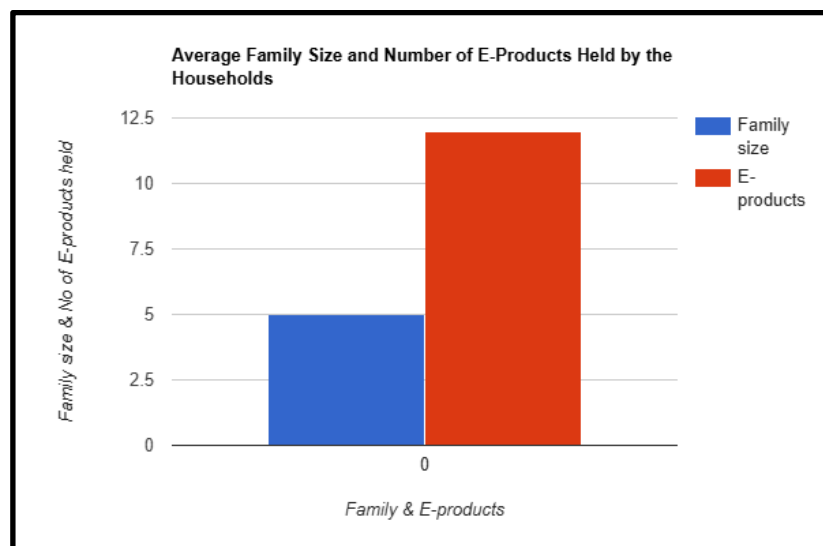
Batteries from discarded electronics, if not handled properly, can lead to **fires in landfills and waste facilities**, posing safety hazards.

Proper **awareness, responsible disposal practices, and improved recycling systems** are essential to reducing these negative effects.

FINDINGS:

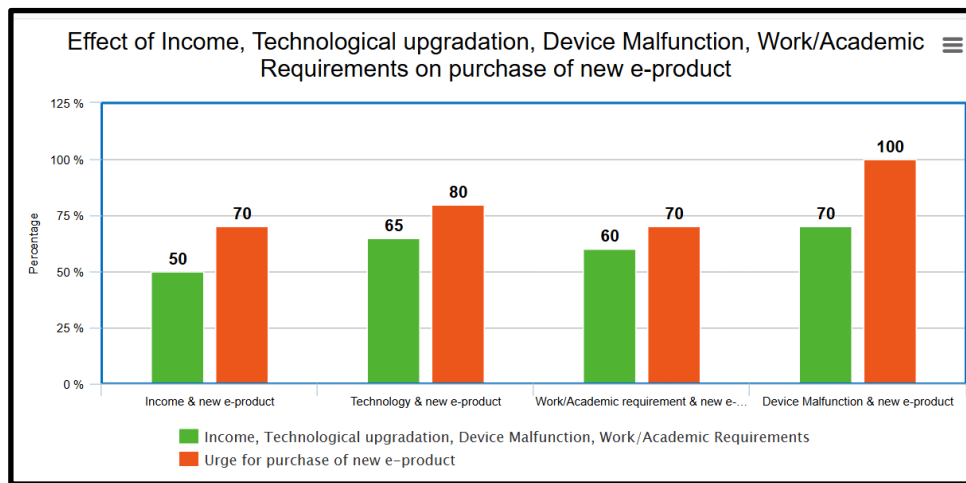
The research study provides valuable insights into e-waste management practices, consumer behavior, and disposal patterns in Chennai. The findings highlight the increasing dependency on electronic products, the challenges in responsible disposal, and the factors influencing e-waste generation and management. Key observations from the survey are presented below.

Figure 1.1



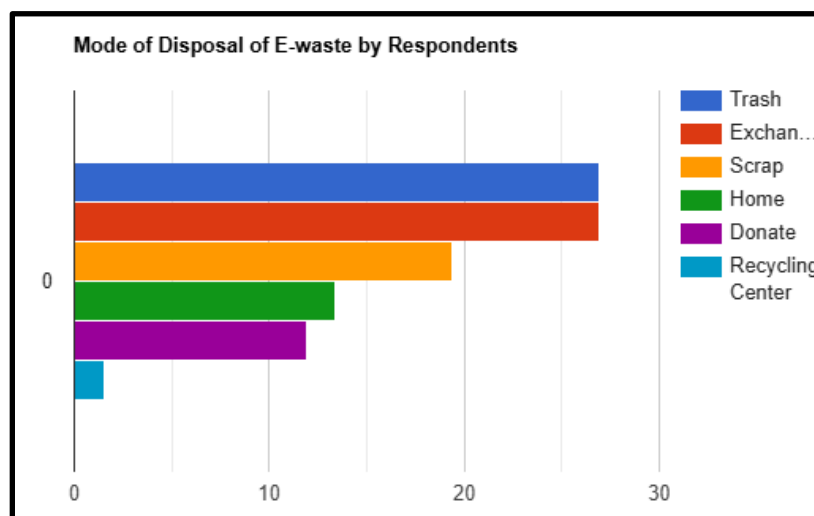
The study reveals that households in Chennai **consume a significant number of electronic products**, highlighting the growing dependence on technology in daily life. The increasing reliance on electronic devices for communication, entertainment, and household activities indicates a rising trend in e-waste generation. Notably, the findings show that the average number of electronic products owned by a household is more than twice the average family size, emphasizing the high rate of electronic consumption. This pattern suggests a pressing need for efficient e-waste management strategies to address the challenges posed by rapid electronic product turnover and disposal.

Figure 1.2



The study also reveals a strong urge to purchase electronic products among the younger generation, particularly those in the **18 to 25 age group**, who constituted the majority of respondents. This growing demand is driven by multiple factors, with key influences being **rising income levels, rapid technological advancements, academic or work-related needs, and device malfunctions**. Additionally, psychological and social factors such as **peer influence, Fear of Missing Out (FOMO), attractive discount sales, EMI options, and the aspiration to own luxury gadgets** further contribute to the increasing purchase of new electronic products. These trends highlight the evolving consumer behavior and its impact on e-waste generation, emphasizing the need for awareness and sustainable disposal practices.

Figure 1.3



The analysis reveals that **most respondents dispose of their e-waste either in regular trash or by selling it as scrap**, indicating widespread improper disposal practices. A notable number of individuals also **store old electronic devices at home**, which eventually lose their exchange value and are later discarded in the same manner. **Only a very small percentage of respondents reported taking their e-waste to nearby recycling**

centers, highlighting a lack of awareness and accessibility to proper recycling facilities. Additionally, many people remain **unaware of the harmful environmental and health impacts of improper e-waste disposal**, underscoring the urgent need for awareness campaigns and improved e-waste management infrastructure.

Important findings:

- The purchase of new electronic products is increasing, driven by **higher income levels, rapid technological advancements, work or academic requirements, and evolving consumer preferences**.
- The purchase of new electronic products is increasing, driven by **higher income levels, rapid technological advancements, work or academic requirements, and evolving consumer preferences**.
- Several factors contribute to the low demand for refurbished electronic products, including:
 - ❖ **Concerns over quality and reliability**.
 - ❖ **Planned obsolescence**, where rapid technological advancements make refurbished products less desirable.
 - ❖ **Limited or no warranty**, discouraging consumer trust.
 - ❖ **Unattractive pricing**, as the cost of refurbished products often falls between the original price and exchange price, leading consumers to save a bit more and opt for a new product instead.
- Trust in second-hand **non-electronic products** is significantly higher compared to electronic products due to:
 - **Ease of repair and maintenance** post-purchase.
 - **Lower investment risk** as these products are typically more affordable.

In contrast, **electronic products are perceived as riskier investments**, as issues may not always be easy to fix, and the cost of purchasing even a second-hand electronic item remains moderate to high for middle-class consumers.

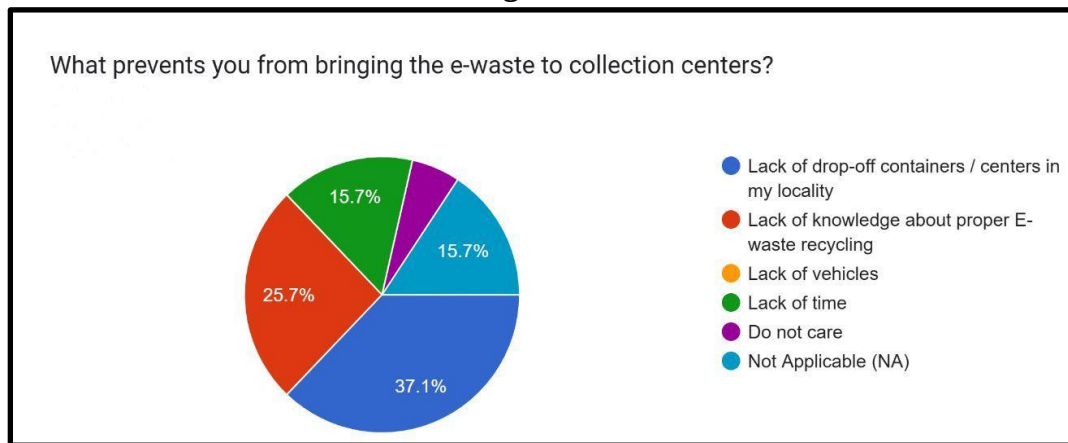
These findings highlight the evolving consumer behavior towards electronic products and the challenges associated with promoting the refurbished electronics market.

- When an electronic product, such as a **mobile phone, gets damaged or malfunctions**, many consumers prefer **disposing of it in regular trash or selling it as scrap** rather than exchanging it for a new one. This behavior is influenced by:
 - **Tedious and Intimidating Exchange Process**: Most online platforms have a **strict evaluation process**, where even minor issues in the device can result in a **significant reduction in exchange value**, making the process frustrating for consumers.

- **Negligible Exchange Value:** Many respondents feel that the offered exchange value is too low to be worthwhile, leading them to opt for **easier disposal methods** rather than going through the hassle of exchange.

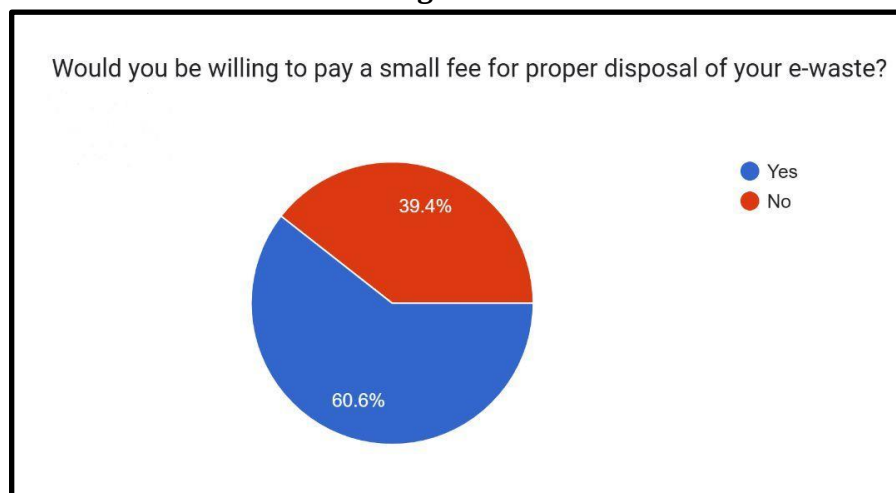
This finding highlights the need for **more consumer-friendly exchange policies** and increased awareness about proper e-waste disposal methods to discourage irresponsible discarding of electronic devices.

Figure 1.4



Several barriers hinder responsible e-waste disposal, primarily due to **lack of drop-off centers, low awareness, and transportation issues**. Many are unaware of proper recycling methods, while others find it inconvenient due to time constraints. A small but notable group remains indifferent to recycling, highlighting the need for **better education and awareness**. Solutions include **more collection centers, mobile pick-up services, and incentive programs** like discounts on new electronics. **Government policies and corporate initiatives** can play a vital role in improving e-waste recycling participation.

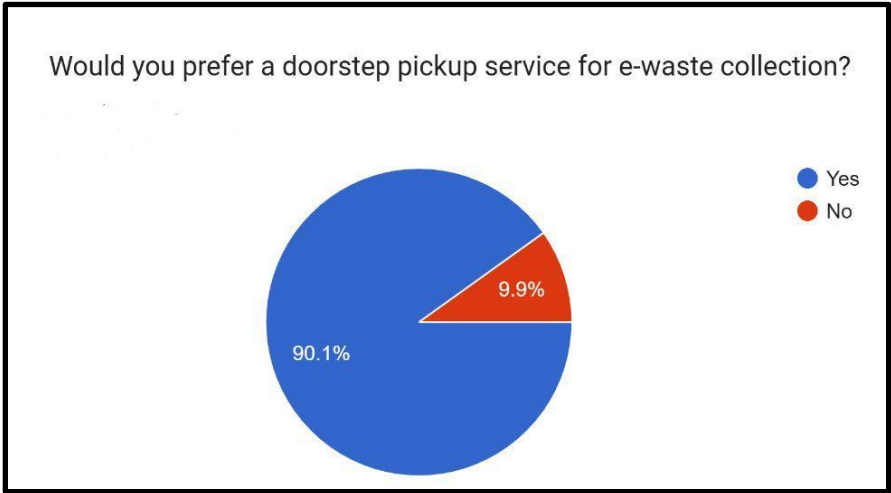
Figure 1.5



The willingness to pay for e-waste disposal reflects public awareness and commitment to sustainability. A **high "Yes" response** suggests recognition of its importance, while a **high "No" response** indicates reluctance, possibly

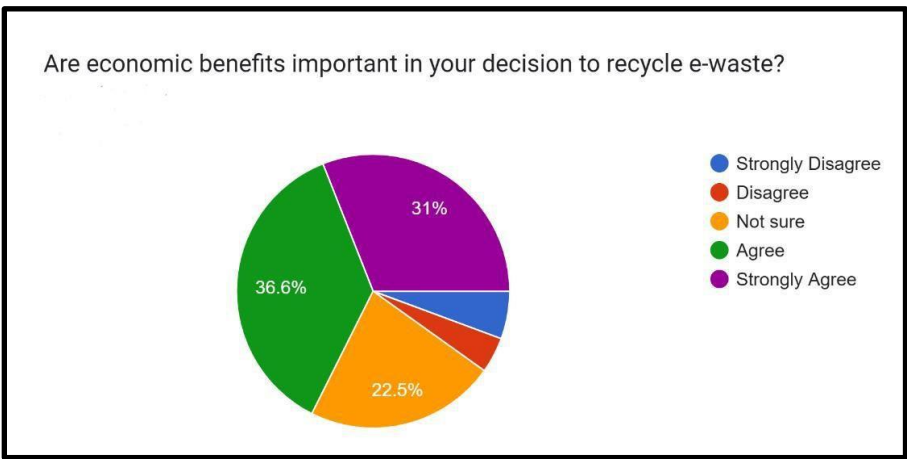
due to the belief that disposal should be free. **Lack of awareness** about environmental impacts may also contribute to this resistance. Governments and organizations can **educate the public, highlight benefits, and introduce incentives** like discounts on new electronics to encourage participation in paid e-waste disposal programs.

Figure 1.6



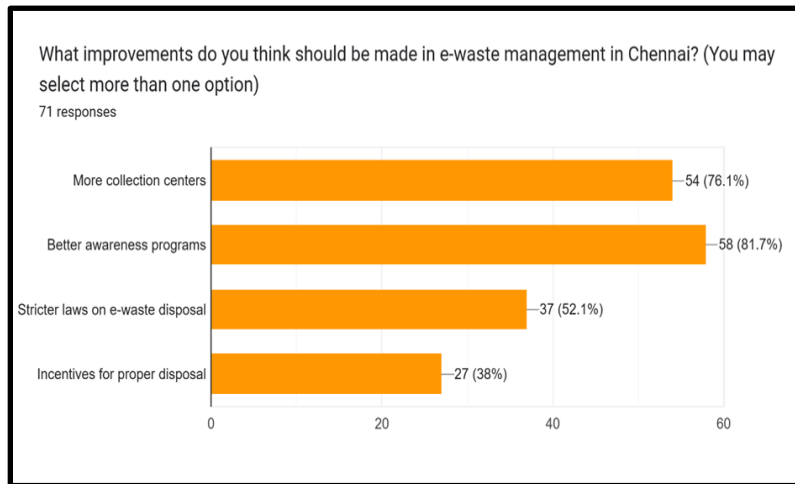
The response to this question gauges interest in **doorstep e-waste collection**. A **high "Yes" response** suggests that accessibility is key to encouraging responsible disposal, as people prefer **convenient recycling options**. A **high "No" response** may indicate **trust issues, privacy concerns, or preference for other methods**. Implementing **secure and reliable** doorstep collection services, especially in urban areas, along with awareness campaigns, can **boost e-waste recycling rates**.

Figure 1.7



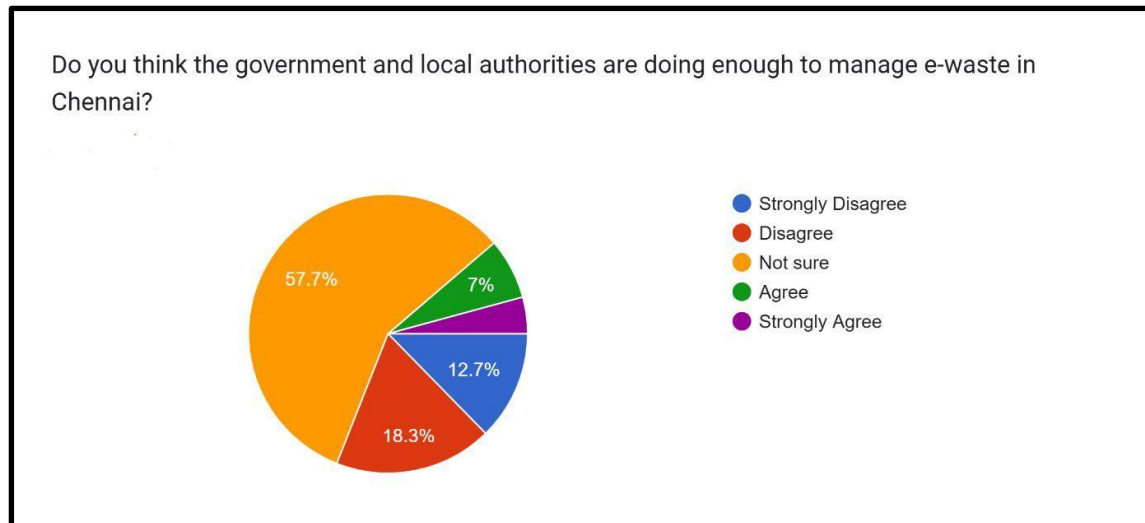
This question assesses the impact of **financial incentives on e-waste recycling**. A **high agreement** suggests that **monetary benefits** like discounts or rebates drive participation, while a **low agreement** indicates **environmental awareness** as the key motivator. **Uncertainty among respondents** highlights the need for better awareness. **Buy-back programs and rewards** can boost recycling rates, making financial incentives a strong tool for **promoting responsible e-waste management**.

Figure 1.8



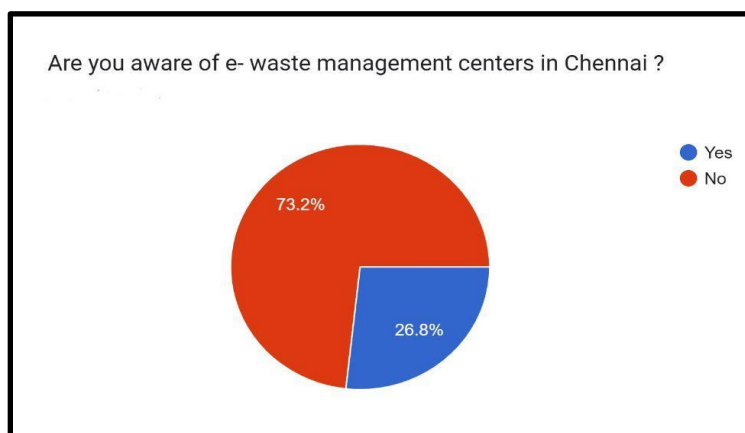
The survey highlights key solutions for **improving e-waste management in Chennai**. **76.1% favor more collection centers**, emphasizing **accessibility** as crucial for responsible disposal. **52.1% support stricter laws**, advocating for **stronger regulations and penalties**. **38% prefer financial incentives**, showing that **discounts and rebates** could encourage participation. A **combined approach of better access, enforcement, and incentives** would be most effective in enhancing e-waste management.

Figure 1.9



Public opinion on **government efforts in e-waste management** is divided. While some find measures sufficient, others see a **gap in policy execution** due to **limited recycling infrastructure and accessibility**. **Lack of visible action** may reduce public participation. To improve engagement, authorities should **expand recycling facilities, enforce stricter regulations, offer incentives, and enhance transparency** through progress reports, fostering **trust and cooperation** in e-waste management.

Figure 1.10



Awareness of **e-waste management centers in Chennai** varies, with many **unaware of proper disposal options**. This **communication gap** leads to **improper disposal** or e-waste accumulation at home. **Stronger outreach** through **public campaigns, social media, and community events** is needed. Providing **accessible information on drop-off locations** can encourage responsible e-waste disposal.

Table 1.1

Income level	Once in a month	More than once in a month	Once in 3 months	Once in 6 months	Yearly once
Less than ₹15,000	NIL	NIL	2.7%	5.4%	5.4%
₹15,001 to ₹30,000	NIL	NIL	1.3%	2.7%	2.7%
₹30,001 to ₹70,000	1.3%	1.3%	6.8%	4.1%	6.8%
₹70,001 to ₹1,00,000	NIL	NIL	1.3%	NIL	4.1%
More than ₹1,00,000	NIL	NIL	1.3%	4.1%	NIL

The table highlights a connection between income levels and participation frequency in a specific activity. Individuals in lower-income brackets (below ₹30,000) participate infrequently, mostly once every three or six months. Those in the middle-income range (₹30,001 – ₹70,000) engage slightly more often, with some taking part monthly and a notable percentage every three or six months. The interpretation suggests a positive correlation between changes in income levels and the frequency of purchasing electronics or gadgets.

Table 1.2

S. No.	Method of e-waste disposal	Primary school	High School	Diploma	UG	PG
1	Throw it in trash	NIL	2.7%	1.3%	13.7%	11%
2	Throw it for scrap	NIL	0%	0%	11%	8.2%
3	Store it at home	NIL	1.3%	0%	11%	2.7%
4	Exchange it	NIL	0%	1.3%	20.5%	4.1%
5	Donate it	NIL	0%	0%	5.5%	6.8%
6	Recycling center	NIL	0%	0%	0%	0%

The data presented in the table highlights the percentage distribution of respondents based on their chosen method of e-waste disposal. The findings clearly indicate that **higher educational qualifications do not necessarily translate into responsible e-waste disposal practices**. Respondents with **undergraduate (UG) or postgraduate (PG) degrees** continue to dispose of e-waste in **regular trash or sell it as scrap**, while some prefer to **store old electronic devices at home** rather than opting for proper recycling methods.

A particularly alarming finding is that **none of the respondents selected ‘taking e-waste to recycling centers’ as their disposal method**, emphasizing a **lack of awareness or accessibility** to proper e-waste management facilities. This evidence supports the null hypothesis that **the method of e-waste disposal (trash, scrap, or recycling) is independent of the consumer’s level of education**, highlighting the need for targeted awareness campaigns and accessible recycling options to promote environmentally responsible behavior.

CONCLUSION:

E-waste management is a critical environmental challenge, particularly in metropolitan cities like **Chennai**, where the consumption of electronic products is rapidly increasing. When not disposed of properly, e-waste releases **toxic substances** that pollute the environment, contaminate soil and water, and pose severe health risks. Effective **e-waste management** is essential to mitigate these impacts and promote **sustainable development**, especially for developing countries like **India**, where urbanization and technological advancements are accelerating electronic consumption.

This research study analyzed the **knowledge, attitude, and practices** of stakeholders regarding e-waste disposal in Chennai. It identified the types of gadgets frequently discarded, assessed the **environmental and health impacts** of improper disposal, and examined current **e-waste generation trends**. The study also tested two hypotheses: **income level does not significantly impact the frequency of purchasing new electronic products**, and **the method of e-waste disposal is independent of the consumer’s education level** – the first one was proven incorrect while the second hypothesis was proven correct.

With the increasing production and consumption of electronic goods, **proper e-waste management is more important than ever**. Chennai, being a major metropolitan city, must implement **stricter regulations, improve public awareness, and enhance recycling infrastructure** to ensure responsible e-waste disposal. Collaborative

efforts from **government agencies, industries, and consumers** are crucial in building a sustainable and environmentally responsible approach to **e-waste management in India**.

RECOMMENDATIONS:

Effective **e-waste management** is crucial for minimizing environmental hazards and promoting sustainable practices, especially in a rapidly growing metropolitan city like **Chennai**. Based on the findings of this research, several key areas require improvement:

- **Implement a producer take-back fee** – Manufacturers and retailers can introduce a **small collection fee** for e-waste disposal, similar to how online platforms charge for picking up exchange products. This will encourage **responsible disposal** and ensure proper recycling.
- **Doorstep collection services** – As **doorstep pickup** is highly preferred by respondents, implementing such services can provide a **safe, secure, and convenient** way for consumers to dispose of their e-waste responsibly.
- **Awareness campaigns and workshops** – Increasing **public awareness** through **campaigns, workshops, and educational programs** will help people understand the **environmental and health hazards** of improper e-waste disposal and encourage better practices.
- **Enhancing refurbished product quality** – Manufacturers should **improve the quality and reliability** of refurbished electronics to **increase consumer trust** and promote a circular economy, reducing e-waste generation.
- **Simplify exchange processes** – Reducing **strict scrutiny** on exchanged electronic products will make it easier for consumers to participate in **exchanging their older e-products for new ones**, preventing improper disposal of old devices.
- **Increase the number of Recycling Centers** – The government and private stakeholders should establish **more e-waste recycling centers** to improve **accessibility** and encourage responsible disposal practices.
- **Monetary incentives for responsible disposal** – The government can **reward consumers** who follow proper e-waste disposal methods by offering **monetary incentives**, such as discounts on new electronics or tax benefits, to drive participation in e-waste recycling initiatives.

By implementing these **recommendations**, Chennai can move toward a **more sustainable and efficient e-waste management system**, reducing environmental harm and promoting responsible electronic waste disposal.

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